Supplementary Material — Software-as-a-Graph: Parameter Sensitivity of the Explanation Layer

Ibrahim Onuralp Yigit[a], Feza Buzluca[a]

[a] *Department of Computer Engineering, Istanbul Technical University, Istanbul, 34469, Turkey*

This supplement reports the parameter-sensitivity analyses supporting RQ3 of the main manuscript. They are placed here rather than in the body because they establish *robustness* of the reported orderings rather than any headline result: their collective finding is that the explanation layer's ten declared weight constants are not load-bearing, with the two exceptions identified below. Section and table numbers are prefixed **S**. All figures are regenerable from the committed artifacts listed in the replication package.

S1. Parameter Sensitivity of the Explanation Layer

The RM composite of Section 5 of the main manuscript declares ten weight constants. We screen all of them, one factor at a time where a natural sweep range exists and jointly via Morris elementary effects [1, 2] otherwise, and report both views in Table S1. Morris is a screening design: μ^* ranks influence on mean ρ without decomposing it, and a σ comparable to a factor's own μ^* signals that its effect depends on where the other nine sit rather than being a fixed marginal slope. It is the computationally efficient alternative to variance-based global analysis [3, 4].

Table S1: Sensitivity of the RM composite to its ten declared weight constants. **OFAT** reports one-factor-at-a-time sweeps against $I^*(v)$ over the seven core synthetic domains; **Morris** reports elementary-effects screening over six scenarios (10 trajectories, 110 evaluations), where μ^* is influence on mean ρ and σ its interaction spread. Rows are ordered by μ^*. Only r_α and λ are load-bearing.

Constant	**One-factor sweep (OFAT)**	**Morris μ^***	**Morris σ**
r_α (Fault Tolerance / Availability blend)	not swept individually	**0.132**	0.040
λ (AHP shrinkage, uniform $\rightarrow$ raw)	$\rho = 0.319 \rightarrow 0.200$, monotone (spread 0.119)	**0.134**	0.090
w_{dur} (QoS durability)	part of the AHP-derived QoS vector $(0.24, 0.62, 0.14)$; not swept separately	0.013	0.016
w_{prio} (QoS priority)		0.022	0.024
w_{rel} (QoS reliability)		0.012	0.018
α (topic payload size)	swept jointly over the full (β, α, ψ) simplex, 7 points: $w(t)$ ordering never falls below $\rho = 0.919$ against the shipped split; downstream spread 0.031 (Topo-QoS), 0.007 (RM)	0.019	0.026
β (topic QoS term)		0.025	0.028
ψ (topic frequency)		0.011	0.011
p (power-mean exponent)	not swept individually	0.005	0.007
γ (library fan-out)	not swept individually	0.001	0.001

Three things follow. First, the parameter risk of the explanation layer is concentrated in two constants, both internal to the RM composite; the QoS-derived weights on which the framework's architectural story rests are not load-bearing for its output. Second, the topic-weight split $(\beta, \alpha, \psi) = (0.75, 0.15, 0.10)$ is a documented convention rather than a tuned parameter — no choice within its simplex, including uniform weighting, would meaningfully change any result in this paper. Third, Dirichlet sampling over 100 draws of the whole weight vector confirms that the parameterization is stable as a whole: mean $\rho = 0.200$ (sd 0.010, 90% interval $[0.185, 0.218]$), with mean Kendall $\tau = 0.826$ and mean top-20% Jaccard 0.739 against the shipped ranking.

The elicited weights are anti-predictive, and we keep them anyway. The λ row deserves separate comment because its direction is unfavorable. Rank correlation falls monotonically as the intra-dimension weights move from a uniform prior toward the raw AHP judgment ($0.319 \to 0.200$, Figure S1): expert elicitation makes the ranking worse, by 0.119. We retain $\lambda = 0.70$ regardless, and the reasoning should be explicit rather than assumed. RM is an attribution instrument, not a ranking model (Section 6.2 of the main manuscript); its purpose is to say *why* a component is fragile in auditable, standards-referenced terms, and rank correlation against a cascade oracle is not the quantity it is built to maximize. Tuning λ toward zero would improve a number we do not claim at the cost of the traceability we do. What this sweep establishes is nonetheless a genuine limitation: we have evidence that the elicited weights produce worse rankings and none that they produce better attributions. Validating the attribution itself — against practitioner judgment, or against the outcome of applying the repairs it recommends — is outside this study and is the explanation layer's principal open question (Section 8.4 of the main manuscript).

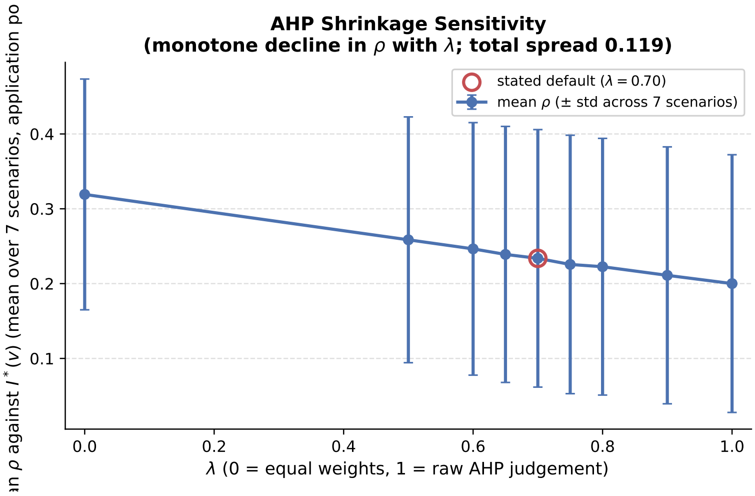

Figure S1: RM composite rank correlation against $I^*(v)$ across the AHP shrinkage parameter λ (Application population, mean over the seven core synthetic domains). The descent is monotone: the closer the intra-dimension weights move to raw elicited judgment, the worse the ranking. Rendered from `results/ahp_shrinkage_sweep_v3.json`.

S2. Zero-Inflation Sensitivity of the Agreement Figures

Spearman ρ over a population where many components are tied at exactly zero is driven substantially by how those ties are handled, and $I^*(v)$ is heavily zero-inflated: 19 to 106 Applications per scenario carry exactly zero cascade impact, whereas $I_{\text{comp}}(v)$ has no exact zeros at all. We therefore report, alongside each full-population figure, $\rho_{>0}$ — the same correlation restricted to components both oracles score strictly positive. It is a sensitivity bound, not a replacement: for a component the simulator actually injected, zero impact is a real measurement ("its failure reaches nobody"), and dropping it would be a results-favorable filter.

The bound changes the reading of two scenarios, in opposite directions. On `hub_and_spoke`, I_{comp} and I^* correlate at $\rho = -0.044$ over the full population but $\rho_{>0} = +0.263$ over the 22 components both score positive: the apparent *sign* disagreement is an artifact of tied zeros, not evidence that the oracles order active components inversely. On `microservices`, the movement runs the other way — $\rho = 0.461$ falls to $\rho_{>0} = 0.257$, so a substantial part of that scenario's agreement is the two oracles concurring on which components are inert rather than on how the active ones rank. Averaged over the seven scenarios the two summaries are close ($\rho = 0.425$, $\rho_{>0} = 0.447$), which is precisely why the per-scenario figures matter: the mean conceals compensating movements of 0.2–0.3 in both directions.

S3. Domain-Specific Weighting and Threshold Sensitivity

Sweeping the composite reliability weight $w_R \in [0, 1]$ moves mean ρ by only 0.018 (0.317 at $w_R = 0$ to 0.336 at $w_R = 1$), and the domain-derived and static rankings agree at mean Kendall $\tau = 0.980$ — the two orderings are very nearly the same ordering. Within that narrow band the domain-derived weighting does not improve on either alternative it replaces: mean ρ is 0.318 (domain-derived) against 0.331 (equal) and 0.321 (static), so it is marginally the *worst* of the three, losing to equal weighting by 0.013 and to static by 0.003. Both margins are far smaller than the between-model differences of Section 7.1 of the main manuscript and we draw no ranking conclusion from either. What the sweep establishes is that no weighting confined to this parameter could move ρ appreciably, so ISO/IEC 25019 context-of-use reweighting must be understood as an attributional device that expresses criticality in stakeholder terms — and not, on this evidence, as a mechanism that makes the ranking more accurate. Two free parameters of the ground truth itself matter more than any scoring weight, and we sweep both. The cascade propagation threshold moves mean ρ across a spread of 0.084 (0.189 at a threshold of 0, rising to 0.271 at 0.5 and flat thereafter at 0.273): a permissive threshold that lets every edge propagate produces a noisier target than one that requires meaningful coupling, and the ordering stabilizes once the cutoff reaches 0.5. Feature normalization spans 0.037 — robust scaling at 0.234 against 0.197 for both min–max and z-score. That is smaller than the threshold spread but of the same order, not negligible beside it: rank-based robust scaling and the two magnitude-preserving alternatives do not induce the same ordering, and the choice is a reported parameter of the scorer rather than an implementation detail. Neither spread indicates that the ordering is *correct*; they indicate only that it is stable under the free parameters of the oracle and the scorer, which is the weaker property a sensitivity sweep can establish.

S4. AHP Pairwise-Comparison Matrices and Their Consistency

Several weight vectors in this framework are derived by the Analytic Hierarchy Process. We state the two that back reported quantities, and we report a property of the matrix family that bears on how much the accompanying consistency ratios are worth.

A caveat on consistency ratios. Saaty's consistency ratio detects *in*consistent judgment; it does not detect a matrix constructed from its own answer. If a priority vector w is chosen first and entry (i, j) is then filled in as w_i/w_j, the resulting matrix is perfectly consistent, $CR \approx 0$, and the statistic certifies nothing. Such a matrix is rank-one: every row is a scalar multiple of every other. Writing $\sigma_1 \geq \sigma_2$ for its two largest singular values, the ratio σ_2/σ_1 separates the two cases — it is near zero for a back-filled matrix and appreciably positive for one carrying genuine judgment disagreement.

Table S2: Consistency diagnostics for the framework's five AHP matrices. CR alone would suggest all five are sound; σ_2/σ_1 shows that three encode a declared priority vector rather than independent pairwise elicitation. The Availability matrix returns a slightly negative CR, which is not attainable for a genuinely elicited matrix ($\lambda_{\max} \geq n$) and arises here as floating-point noise around exact consistency.

Matrix	n	CR	σ_2/σ_1
Topic QoS (main manuscript, Eq. 3)	3	+0.0158	0.072
Fault Tolerance	3	+0.0029	0.027
Composite impact (I_{comp})	4	+0.0011	0.014
Maintainability	5	+0.0005	0.0006
Availability	5	−0.0029	0.0006

Only the Topic QoS matrix carries a consistency ratio that means what a consistency ratio is supposed to mean, and it is the one matrix a continuous-integration test guards on both sides: `tests/test_ahp_shrinkage.py` asserts $0.005 < CR < 0.10$, the lower bound existing precisely as a non-degeneracy check. We report the remaining vectors as declared constants with documented structure rather than as elicited judgment, and note that this reading costs the framework little: the global sensitivity analysis (Section S1) finds eight of the ten weight constants to have $\mu^* \leq 0.023$, so the ranking results in Section 7 of the main manuscript do not rest on these values being optimal.

Table S3: Saaty matrix over the three Topic QoS dimensions backing QoS(t) in Equation (3) of the main manuscript. Geometric-mean priority vector $(0.2385, 0.6250, 0.1365)$; $\lambda_{\max} = 3.018$, $CI = 0.0092$, $RI = 0.58$, $CR = 0.0158$. The shipped constants $(0.24, 0.62, 0.14)$ round this within 0.005.

	Reliability	**Durability**	**Priority**
Reliability	1	1/3	2
Durability	3	1	4
Priority	1/2	1/4	1

Table S4: Matrix over the four composite-impact criteria — reachability loss (RL), fragmentation (FR), throughput loss (TL), flow disruption (FD) — backing $I_{\text{comp}}(v)$ (Section 4.3 of the main manuscript). Raw priority vector $(0.389, 0.255, 0.255, 0.100)$; after the framework's $\lambda = 0.7$ shrinkage toward a uniform prior, $(0.347, 0.254, 0.254, 0.145)$, which the shipped $(0.35, 0.25, 0.25, 0.15)$ round. Per Table S2 this matrix is rank-one, so it documents the provenance of those constants without independently justifying them.

	RL	**FR**	**TL**	**FD**
RL	1	3/2	3/2	4
FR	2/3	1	1	5/2
TL	2/3	1	1	5/2
FD	1/4	2/5	2/5	1

S5. Generative Parameters of the Synthetic Corpus

Table S5 records the configuration behind each synthetic scenario of Table 4 in the main manuscript. It is replication detail rather than a result: the corpus shape a reader needs to interpret the evaluation is in Table 4 itself, and every configuration is committed in the replication package.

Table S5: Generative parameters of the twelve synthetic scenarios (eleven evaluation scenarios plus the ATM case study). Seed and fan-out figures are read from the committed configurations; per-entity counts are in Table 4 of the main manuscript. The modal QoS column gives the most common reliability/durability/priority value and the range of topic shares carrying them, computed from the committed topology rather than the config's declared QoS targets, which domain-driven assignment does not always realize (Section 6.1 of the main manuscript).

Scenario	**Seed**	**Pub**	**Sub**	**Modal QoS (R/D/P)**
Air Traffic Management (ATM)	42	1.2	2.9	RELIABLE/VOLATILE/HIGH (41–81%)
Autonomous Vehicle (AV)	1001	2.5	5.0	RELIABLE/TRANSIENT_LOCAL/HIGH (45–80%)
Enterprise Pub-Sub	7007	3.0	4.5	RELIABLE/TRANSIENT_LOCAL/MEDIUM (29–79%)
Financial Trading	3003	4.0	6.0	RELIABLE/PERSISTENT/HIGH (40–89%)
Healthcare Integration	4004	2.5	3.0	RELIABLE/PERSISTENT/HIGH (40–88%)
Hub-and-Spoke	5005	2.0	7.0	RELIABLE/TRANSIENT_LOCAL/MEDIUM (33–67%)
Industrial SCADA	1902	1.1	1.0	RELIABLE/TRANSIENT/HIGH (31–80%)
IoT Smart City	2002	2.0	1.5	BEST_EFFORT/VOLATILE/LOW (56–75%)
Logistics Fleet	2104	1.8	1.9	RELIABLE/TRANSIENT_LOCAL/MEDIUM (40–76%)
Microservices Mesh	6006	1.5	2.0	RELIABLE/TRANSIENT_LOCAL/MEDIUM (40–60%)
Real-Time Gaming	2003	2.6	2.8	BEST_EFFORT/VOLATILE/HIGH (39–71%)
Telecom RAN	1801	2.2	1.6	RELIABLE/VOLATILE/HIGH (36–67%)

S6. Anti-Pattern Detection and Node-Type Stratification

Validated against $I_{\text{comp}}(v)$, the rule-based anti-pattern catalog reaches mean precision 0.244 and recall 0.915 ($F_1 = 0.386$) — but it implicates 93.8% of scored components, so that recall is close to what indiscriminate flagging achieves and the precision is near the base rate. The ATM scaling sweep (29 to 444 components, five seeds, seed-invariant) shows the mechanism: Cohen's κ ends at -0.018 on the 444-component instance, against $+0.035$ at 29 components, with a flag rate of 92.3% at the largest scale. The path is not monotone — κ rises to $+0.086$ at 148 components before collapsing — so we do not claim a clean decay law, only the endpoint: at the largest scale the catalog's agreement with the oracle is indistinguishable from chance, and marginally on the wrong side of it. The catalog does not become wrong at scale so much as it stops discriminating. We report it as a characterization of the catalog rather than a working triage mechanism, and delegate critical-set identification to the continuous rankers of Sections 7.1–7.2 of the main manuscript. A nineteenth detector, `DEEP_PIPELINE`, is excluded for path-combinatorial explosion (247,761 paths on a 29-component fixture).

Stratification vs. Pooling: Measured against the composite oracle $I_{\text{comp}}(v)$ over the eight scenarios of the detection benchmark, stratified RM rank correlations are $\rho = 0.566$ (Application, 8 scenarios), 0.119 (Broker, 6 scenarios), and 0.244 (Node, 8 scenarios), while pooled correlation across all types collapses to $\rho = 0.098$. Pooling node types triggers Simpson's paradox by conflating disparate structural base rates across architectural layers: the pooled figure sits below every per-type figure it aggregates. This is why every evaluation in this paper is reported on a single stratum, and why pooled critical-set numbers should be read as inflated wherever they appear (Section 7.4 of the main manuscript). On that same pooled benchmark, unweighted degree centrality attains $\rho = 0.168$ and $F_1 = 0.442$ against RM's 0.098 and 0.342 — RM is beaten by degree on both, confirming that it serves as an attribution instrument rather than an unconstrained global ranker.

S7. Real-World Evaluation of the Explanation Layer

We evaluated the framework on five open-source distributed systems transcribed from their public repositories by dedicated architectural adapters, with results presented in Table S6. We note the scope of this evaluation before reporting it. Online Boutique is a vendor demonstration application and Train-Ticket an academic benchmark, Home Assistant is an authentic open-source IoT automation platform, and EdgeX Foundry is an industrial edge computing reference implementation; each is an abstraction of its upstream repository rather than a complete transcription; and their labels come from the same simulation oracle used throughout, so what follows tests topological generalization, not agreement with observed field failures. Two further limits bound what this table can support. First, the predictor scored in Table S6 is the deterministic explanation layer $Q(v)$ together with the two training-free topological baselines, and it is scored against $I_{\text{comp}}(v)$; the learned model is evaluated separately in Section 7.4.1 of the main manuscript against $I^*(v)$, and the two are not commensurable. Second, the labels are strongly tied at zero in several architectures — 13 of 32 Applications in Autoware, 14 of 22 in Cloud Microservices, 27 of 41 in Train-Ticket, and 12 of 22 in EdgeX carry zero simulated impact (the complement of the "Impactful Apps" column), whereas Home Assistant exhibits much lower zero-inflation with 17 of 24 applications actively participating in failure cascades. Spearman ρ over populations that are heavily tied is dominated by how those ties are handled; the values below use midrank ties throughout.

Key Insights for RQ4:

1. **The explanation layer ranks unseen architectures well, but this is not a learned-transfer result.** RM / $Q(v)$ attains $\rho = 0.800$ on EdgeX Foundry, 0.778 on Cloud Microservices, 0.759 on Train-Ticket, 0.685 on Autoware.universe, and 0.514 on Home Assistant. Because $Q(v)$ is a closed-form scoring function rather than a fitted model, applying it to a new architecture involves no transfer in the inductive sense of Section 7.2 of the main manuscript: nothing was trained, and no distribution shift is being survived. What this table establishes is that the deterministic attribution of Section 5 of the main manuscript remains informative on architectures we did not generate. The learned model is a separate question, answered in Section 7.4.1 of the main manuscript.

Table S6: Open-source reference systems scored by the deterministic explanation layer RM / $Q(v)$ — no GNN checkpoint is invoked, so $\pm$ is oracle variance over five simulation seeds, not training variance. "Gain vs. Deg" is $|\rho_Q| - |\rho_{\text{deg}}|$. Ground truth here is $I_{\text{comp}}(v)$, whereas Table 9b of the main manuscript uses $I^*(v)$: **the two must not be read against each other**. Simulator-derived labels, not observed incidents.

Real-World Architecture	$\|V\|$	$\|V_{\text{app}}\|$	Spearman ρ	Kendall τ	App $F_1@K$	Pooled $F_1@K$	Impactful Apps	Gain vs. Deg
Autoware.universe (ROS 2)	75	32	**0.685 ± 0.010**	0.513	0.333	0.800	19 / 32	+0.357
Cloud Microservices Mesh	60	22	**0.778 ± 0.001**	0.639	0.500	1.000	8 / 22	+0.014
Train-Ticket Booking Mesh	90	41	**0.759 ± 0.001**	0.605	0.625	1.000	14 / 41	+0.264
Home Assistant (Smart Home)	63	24	**0.514 ± 0.016**	0.373	0.250	0.667	17 / 24	+0.289
EdgeX Foundry (Industrial IoT)	63	22	**0.800 ± 0.015**	0.563	0.000	1.000	10 / 22	+0.427

Table S6 is not commensurable with Table 7's RM column ($\rho = 0.133$), and the gap has two separate causes that should not be conflated. The larger is the oracle: Table S6 scores $Q(v)$ against $I_{\text{comp}}(v)$ (`FailureSimulator`), whereas Table 7 of the main manuscript scores it against $I^*(v)$ (`FaultInjector`), and the two agree at only $\rho = 0.425$ (Section 7.3 of the main manuscript). Scored against $I^*(v)$ on these same five systems, RM attains $\rho = 0.516$ (Table 9b of the main manuscript) — still far above its synthetic-corpus figure, so the remainder is a real-versus-generated difference, with the tied-label structure described above contributing to both. The practical consequence is that no RM figure in this paper should be compared across tables without checking which oracle produced it.

2. **Stratified vs. Pooled Critical Set Identification:** On the stratified Application population (V_{app} with $K = \text{round}(0.20 \cdot |V_{\text{app}}|)$), $Q(v)$ identifies the top-20% critical services with $F_1@K = 0.333$ in Autoware, 0.500 in Cloud Microservices, 0.625 in Train-Ticket, 0.250 in Home Assistant, and 0.000 in EdgeX Foundry (where failure impact is dispersed broadly across active device services). When evaluated across the pooled multigraph ($K = \text{round}(0.20 \cdot |V|)$), the pooled $F_1@K$ reaches 0.800, 1.000, 1.000, 0.667, and 1.000, respectively. This disparity is the base-rate phenomenon of Section 6.3 of the main manuscript: the passive infrastructure entities (Topics, Nodes, Libraries) carry constant zero failure impact under the injector configuration used for these runs, so a pooled top-K set is largely populated by correctly identifying inert nodes. The pooled column is therefore inflated and we do not read it; the stratified V_{app} column is the operational figure.
3. **Framework Acceptance Gates and Cross-Domain Disparities:** The framework ships a five-condition release gate ($\rho \geq 0.75$, $F_1 \geq 0.65$, SPOF $F_1 \geq 0.60$, fault-tolerance ratio ≤ 0.30, prediction gain ≥ 0.02). EdgeX Foundry passes all five gates across all five seeds (100% pass rate), driven by strong rank correlation ($\rho = 0.800$), pooled $F_1 = 1.00$, and robust articulation detection (SPOF $F_1 = 0.80$). In contrast, the pass rate is zero on the other four architectures: Autoware fails the ρ and SPOF conditions, Cloud Microservices fails SPOF and prediction gain, Train-Ticket fails SPOF, and Home Assistant misses the ρ gate ($\rho = 0.514$) despite scoring a perfect SPOF $F_1 = 1.00$. What this establishes is a negative result about thresholds, not a positive one about ordering: a gate calibrated on our synthetic corpus fires correctly on one architecture in five, so its absolute cut-offs are domain-specific and do not transfer as shipped. The ordering half of the question — whether the ranking underneath those thresholds transfers — is not answered here and is answered negatively in Section 7.4.1 of the main manuscript, once the population is restricted to components that actually propagate a failure.
4. **Predictive Advantage over Structural Heuristics:** The "Gain vs. Deg" column reports $Q(v)$'s rank-correlation margin over an unweighted degree centrality baseline (+0.427 on EdgeX Foundry, +0.357 on Autoware, +0.289 on Home Assistant, +0.264 on Train-Ticket, +0.014 on Cloud Microservices), so on this population and this oracle the hierarchical attribution of Section 5 of the main manuscript orders components better than degree centrality does. The margin does not generalize beyond those conditions, and we scope it accordingly: on the pooled synthetic detection benchmark of Section 7.3.6 of the main manuscript, measured against the same $I_{\text{comp}}(v)$ but across all entity types, degree *beats* RM on both rank correlation (0.168 vs. 0.098) and $F_1@K$ (0.442 vs. 0.342). The two findings are not in conflict — one is stratified on Applications in five transcribed systems, the other pooled across types in eight generated ones — but neither supports a general claim that RM dominates degree. Wilcoxon signed-rank tests over the five systems do not reach significance ($p \geq 0.33$), which at $n = 5$ they could

hardly do; we report them for completeness rather than as support.

5. **The QoS-weighted heuristic cannot be scored here, and the reason is structural rather than incidental.** The unweighted `Topo` reference is computable ($\rho = 0.307/0.891/0.528$ on Autoware, Cloud Microservices and Train-Ticket; mean 0.511, Table 9b of the main manuscript); `Topo-QoS` is not. The obstacle is not missing QoS data — every topic in all five adapters carries declared **`durability`**, **`reliability`** and **`transport_priority`** profiles, and projecting $w(t)$ onto the structural edges yields non-unit weights on 45.3–62.2% of them. It is that `Topo` reads betweenness off the cached `DEPENDS_ON` projection, whereas the QoS-weighted variant must recompute it on the raw multigraph, where Applications never route messages and betweenness is identically zero for all of them — the same degeneracy Section 6.2.1 of the main manuscript gives as the reason topological baselines are run on the projection at all. We record the gap rather than substitute a number for it; the consequence, that the real-world margin is measured against the weaker training-free reference, is stated in Section 8.4 of the main manuscript. Separately, projection-based baselines assign constant zero to Applications carrying no derived `DEPENDS_ON` edge, accentuating sensitivity to tied ground truth.

S8. HGT Relational Attention Weight Analysis

Figure S2 visualizes the relational mutual attention distributions of a Heterogeneous Graph Transformer on the ATM case study architecture. *The model is not the predictor evaluated elsewhere in this paper.* It is a three-layer, four-head HGT trained for 100 epochs on the ATM scenario alone at seed 42, built by `reproduce/extract_attention.py` for this figure and discarded afterwards. The LOSO-evaluated checkpoints cannot stand in: they are trained over the full corpus, carry a 40-relation schema against this topology's 32, include `Library` publish and subscribe relations the ATM graph does not contain, and are two layers deep rather than three. Nothing below therefore speaks to what the evaluated predictor attends to; it speaks to whether typed attention differentiates at all.

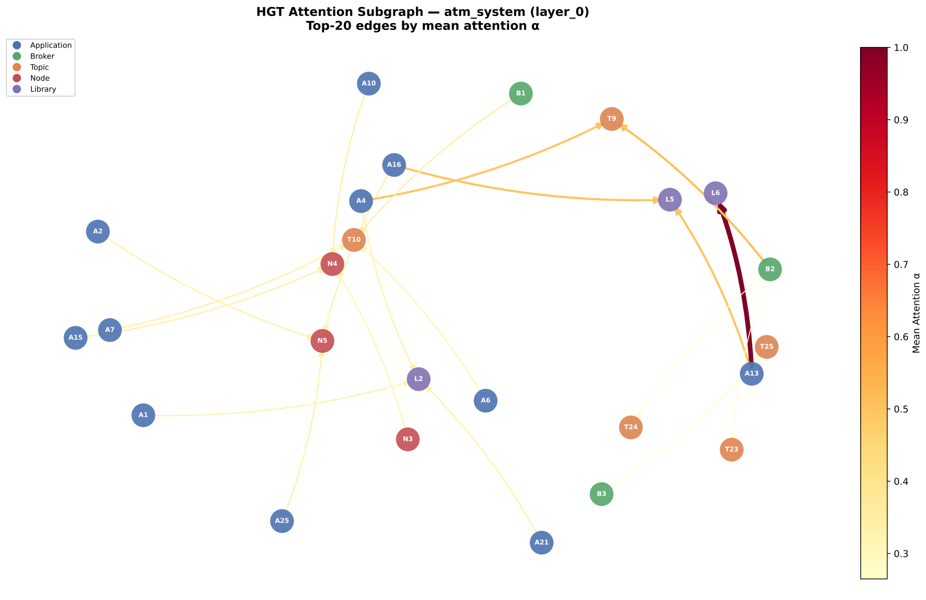

Figure S2: Relational attention on the ATM case study, from a three-layer HGT trained for 100 epochs on this scenario alone (seed 42) — not the predictor evaluated in Sections 7.1–7.4 of the main manuscript. Mean α spans 0.113–0.222 across the eight relation types and is governed substantially by destination in-degree. A qualitative illustration on one topology, not evidence of general relation prioritization.

Aggregated by relation type, first-layer mean attention across all eight relations runs: `USES` (App→Lib) 0.220; `ROUTES` 0.187; `SUBSCRIBES_TO` 0.172; `USES` (Lib→Lib) 0.169; `PUBLISHES_TO` 0.160; `RUNS_ON`

(App→Node) 0.156; `CONNECTS_TO` 0.135; and `RUNS_ON` (Broker→Node) 0.113. Across the three layers no relation-type mean moves by more than 0.019, but the rank order is *not* stable: the four relations between 0.15 and 0.18 are spaced more closely than that drift, and they permute between layers. Three methodological cautions bound what this visualization supports:

1. The spread across all eight relation types is narrow (0.113–0.222, and only 0.156–0.220 among the five carrying more than nine edges), indicating a mild distributional tendency rather than a clear-cut structural separation.
2. Because α is computed as a softmax over each destination node's incoming neighborhood $\mathcal{N}(v)$, the mean attention weight α is substantially governed by local in-degree. The single largest weight in the whole extraction ($\alpha_{uv} = 1.00$, on a `USES` edge into a library) lands on a destination of in-degree one, where $\alpha = 1.0$ holds by arithmetic and carries no information. The thinnest relations sit at the extremes of the ordering: `USES` (Lib→Lib) carries 3 edges and `RUNS_ON` (Broker→Node) carries 5, against 85 for `SUBSCRIBES_TO`.
3. The figure is a function of the corpus snapshot, not a stable property of the architecture. Regenerating the ATM topology changed its edge set without changing any entity count (`PUBLISHES_TO` 56 → 48, `SUBSCRIBES_TO` 90 → 85, `ROUTES` 36 → 37, `CONNECTS_TO` 7 → 9), and that alone moved `USES` (Lib→Lib) from 0.225 to 0.169 — from above `ROUTES` to below it. The extraction itself is deterministic, so the whole of that shift is input sensitivity on a 74-node graph.

Consequently, Figure S2 serves as an illustrative artifact confirming that multi-head heterogeneous attention remains active across relation types, rather than a statistical proof of general relational importance.

References

[1] M. D. Morris, Factorial sampling plans for preliminary computational experiments, Technometrics 33 (2) (1991) 161–174.

[2] F. Campolongo, J. Cariboni, A. Saltelli, An effective screening design for sensitivity analysis of large models, Environmental Modelling & Software 22 (10) (2007) 1509–1518.

[3] A. Saltelli, M. Ratto, T. Andres, F. Campolongo, J. Cariboni, D. Gatelli, M. Saisana, S. Tarantola, Global Sensitivity Analysis: The Primer, John Wiley & Sons, 2008.

[4] I. M. Sobol', Sensitivity estimates for nonlinear mathematical models, Mathematical Modelling and Computational Experiments 1 (4) (1993) 407–414.